MAGNUS DIERKING

Born in Germany, 21 September 1998

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WORK EXPERIENCE

since 05/2026 PhD Student & Research Associate

PEARL *Computer Science Dept.* · *Technical University of Darmstadt*
Research in Robot Learning at the Interactive Robot Perception & Learning (PEARL) Group, supervised by Prof. Georgia Chalvatzaki.

10/2024–05/2025 Research Intern

Huawei R&D *Huawei Technologies Research & Development UK Limited* · *London, Shenzhen*
Extracurricular internship during Master studies. Working on reinforcement learning, imitation learning and supporting middleware for robotics as part of the Embodied AI team.

2023–2024, 2025–2026 Research Assistant

IAS *Intelligent Autonomous Systems Group* · *Technical University of Darmstadt*
Developed a Python/C++ software stack for FR3 robotic arms, built on ROS2 Jazzy.
Assited Ph.D. researchers with experiments, visualization and studies on supporting theory in optimal transport, Bayesian inference and optimization for robot learning.

2018–2021, 2022–2023 Working Student

mas *Medical Airport Service GmbH* · *Frankfurt*
Support personel for the rescue station on the apron of Frankfurt International Airport.

2021–2022 Teaching Assistant

TEMF *Institute for Accelerator Science and Electromagnetic Fields* · *Technical University of Darmstadt*
Developed lesson plans, theoretical / programming tutorials and provided mentorship and feedback for undergraduate students.
Courses: Electrodynamics, Numerics for Electromagnetic Field Simulation

PUBLICATIONS

2025 Ark

arXiv Open-source, Python-first robotics framework that provides a Gym-style interface for collecting data, training policies, and switching seamlessly between simulation and real-robot deployment. It includes reusable modules for control, SLAM, motion planning and visualization, and integrates natively with ROS to accelerate end-to-end robotics research.
First Author

2025 OpenPyro-A1

IEEE RA-L Open-source, low-cost bimanual half-humanoid robot designed for advanced manipulation research. It features a modular, repairable hardware design and supports coordinated two-handed tasks such as folding, cutting, and assembling. The platform enables teleoperation via a Meta Quest 3 and provides interfaces for learning-based controllers to support scalable, real-world robotics experimentation.
Co-Author

EDUCATION

10/2022–04/2026 **Computational Engineering**

Master of Science

GPA: 1.00 · Top of Class · Technical University of Darmstadt

Thesis: *Domain Randomized Deployment of Massively Parallel MPC*
Developed real-to-sim-to-real pipeline for massively parallel SMPC, implementing JAX-based planning stack with GPU-accelerated MuJoCo MJX and ROS2 control on Franka Research 3 for contact-rich manipulation and simulation studies. Evaluated algorithms up to 10 Hz with robust sim-to-real transfer. Analyzed online domain randomization limits, favoring contact-initiation parameters.
Advisor: Prof. Jan Peters · Supervision: Dr. João Carvalho, Dr. An Thai Le · *Grade: 1.00*

Robot Learning · Reinforcement Learning · Generative Models · Robotic Manipulation · Parallel Programming
Differential & Riemannian Geometry · Lie Theory · Information Theory · Numerics · Geometric Algebra · Convex Optimization · Graph-based Learning & Signal Processing · Control Theory · Optimal Transport

10/2019–10/2022 **Computational Engineering**

Bachelor of Science

GPA: 1.51 · Top 10% · Technical University of Darmstadt

Thesis: *Parallel Solution of Linear Systems Arising in Domain Decomposition Methods*
C++ implementation of a parallel solver for large-scale surface PDEs within an Isogeometric Analysis (IGA) framework.
Advisor: Prof. Sebastian Schöps · Supervision: MSc. Maximilian Nolte · *Grade: 1.00*

Robotics · Software Engineering · Algorithms and Data Structures · Functional & Object-oriented Programming · Geometric Modelling
Linear Algebra · Calculus · ODEs & PDEs · Numerics · Statistics & Probability Theory · Classical Mechanics · Signal Processing · Electrodynamics

10/2018–10/2019 **Mechanical Engineering**

Completed Coursework · Technical University of Kaiserslautern

Material Science 1-2 · Mechanical Design Project · Production Engineering · Business Organization

2018 **Elisabeth-Langgässer Gymnasium**

A-Levels Math · English · History

TECHNICAL SKILLS

OS Linux · Mostly Ubuntu, some Arch

Programming Python · JAX, PyTorch, CVX, NumPy, SciPy, Pandas
C++ · Eigen, OpenMP

Simulation MuJoCo, MJX and MJWarp, PyBullet

Tools Git, ROS2, Docker, LaTeX

Hardware Franka Research 3, Trossen Viper, OpenPyro-A1
OptiTrack, Intel RealSense

OTHER INFORMATION

Scholarships
2024 · Erasmus Placements Program
2023 · Deutschlandstipendium

Languages
German · Mothertongue
English · Fluent
French · Basics